

REQUIREMENTS ENGINEERING – ELICITATION

Requirement Gathering Report

Selected Software Project: QR-Based Restaurant Ordering System

1. Introduction

Requirements elicitation is the process of identifying and understanding the needs, expectations, and requirements of the people who will use or be affected by a software system. For the selected QR-Based Restaurant Ordering System, elicitation focuses on understanding the requirements of customers and restaurant staff and translating those needs into clear system requirements.

The selected project allows customers to scan a QR code placed on their table, view a digital menu, select food items, add them to a cart, and place orders using their smartphones. Restaurant staff can manage orders digitally. The project therefore requires requirements from both the customer and restaurant operational perspectives.

2. Stakeholder Identification

Stakeholders are individuals or groups who interact with the system, use its outputs, or are affected by its operation. The major stakeholders identified for the project are:

Stakeholder	Role / Interest	Key Requirements
Customers	Primary users who place food orders.	Easy QR scanning, digital menu, cart, order placement, and order-status visibility.
Restaurant Staff	Receive, prepare, and manage customer orders.	View incoming orders, update order status, and manage completed orders.
Restaurant Management	Interested in efficient restaurant operations and reliable order management.	Accurate orders, organized workflow, reduced manual errors, and maintainable system.
Project Team	Develops, tests, documents, and maintains the system.	Clear requirements, modular design, testability, and maintainability.

3. Requirement Elicitation Techniques

The following suitable elicitation techniques are used as the basis for gathering requirements for the selected system. These techniques help capture requirements from different stakeholder viewpoints.

3.1 Interviews

Interviews can be conducted with restaurant staff and management to understand the current ordering process, common difficulties, order-management needs, and expectations from a digital system.

Sample Interview Questions:

- How are customer orders currently received and recorded?
- What problems occur during busy restaurant hours?
- What information should staff see when a new order arrives?

- Which order-status updates are useful to restaurant staff?
- What difficulties should the digital ordering system reduce?

3.2 Questionnaires

A short questionnaire can be used to gather feedback from a larger number of customers. It can identify customer expectations regarding menu access, ordering speed, ease of use, and order status.

Sample Questionnaire Questions:

- Is scanning a QR code to access a restaurant menu convenient for you?
- Would you prefer placing an order directly from your smartphone?
- Which information would you expect to see on a digital menu?
- How important is it to see your order status?
- What would make a restaurant ordering application easy to use?

3.3 Observation

Observation can be used to study the traditional restaurant ordering process. The observer can examine how customers receive menus, how orders are communicated to staff and the kitchen, and where delays or communication errors occur. These observations help identify practical requirements for the proposed system.

3.4 User Personas

User personas represent typical users of the system and help the development team understand their goals and expectations.

Persona	Goal	Pain Point
Customer	Order food quickly and conveniently.	Waiting for staff and possible ordering delays.
Restaurant Staff	Receive and manage orders efficiently.	Handling multiple manual orders during busy periods.
Restaurant Manager	Maintain an organized and reliable ordering process.	Manual errors and inefficient order coordination.

4. Gathered Functional Requirements

The following functional requirements are derived from the intended workflow of the QR-Based Restaurant Ordering System:

- The system shall identify the restaurant table through its QR code.
- The system shall allow customers to access the digital menu after scanning the QR code.
- The system shall display available food items to customers.
- The system shall allow customers to select food items.
- The system shall allow customers to add food items to a cart.
- The system shall allow customers to remove items from the cart.

- The system shall allow customers to place a valid order.
- The system shall store order information in the database.
- The system shall allow restaurant staff to view incoming orders.
- The system shall allow staff to update order status.
- The system shall prevent an order from being placed when the cart is empty.
- The system shall allow customers to see updated order status.

5. Non-Functional Requirements

- **Usability:** The customer ordering process should be simple and easy to understand.
- **Performance:** Menu pages and order requests should respond within a reasonable time under normal usage.
- **Reliability:** Valid orders should be stored correctly and should not be lost during normal operation.
- **Maintainability:** The system should be organized into separate modules so that individual components can be modified easily.
- **Scalability:** The design should allow future additions such as online payment, notifications, and multiple restaurants.
- **Security:** User input and order information should be validated and handled appropriately.

6. User Stories

User stories describe system requirements from the perspective of the user. The following user stories represent the core requirements of the selected project.

ID	User Story
US-01	As a customer, I want to scan the QR code on my table so that I can access the restaurant's digital menu.
US-02	As a customer, I want to view the digital menu so that I can see the available food items.
US-03	As a customer, I want to select food items so that I can choose what I want to order.
US-04	As a customer, I want to add items to a cart so that I can review my order before placing it.
US-05	As a customer, I want to remove items from my cart so that I can correct my selection.
US-06	As a customer, I want to place my order so that the restaurant can prepare my selected food.
US-07	As a customer, I want to view my order status so that I know whether my order is being prepared or completed.
US-08	As restaurant staff, I want to view incoming orders so that I can prepare them correctly.
US-09	As restaurant staff, I want to update an order's status so that the order progress is recorded digitally.
US-10	As restaurant staff, I want completed orders to be identifiable so that I can manage the order workflow efficiently.

7. Requirement Gathering Report

The requirement gathering activity identifies the major stakeholders, their needs, and the expected system behavior. Customers require a quick and convenient method for accessing the menu and placing orders. Restaurant staff require a clear digital interface for receiving and managing orders. Restaurant management requires an organized process that reduces manual errors and improves operational efficiency.

The gathered requirements establish the core system flow: **Scan Table QR Code → Digital Menu → Select Food → Add to Cart → Place Order → Backend API → Database → Restaurant Dashboard → Order Preparation → Order Completed.**

The requirements also support the project's software engineering goals. The system can be divided into frontend, backend, database, and testing modules, while user interface, business logic, and database operations remain separated. This supports modularity, separation of concerns, maintainability, and testing.

8. Requirement Prioritization

Priority	Requirements
High	QR/table identification, digital menu, food selection, cart, order placement, order storage, staff order viewing and status updates.
Medium	Improved order-status visibility and additional validation/error handling.
Future	Online payment, notifications, and multiple-restaurant support.

9. Conclusion

The requirements elicitation process provides a clear understanding of what the QR-Based Restaurant Ordering System should accomplish. The main stakeholders are customers, restaurant staff, restaurant management, and the project team. Interviews, questionnaires, observation, and user personas are suitable techniques for identifying their needs.

The resulting functional requirements and user stories define the core ordering workflow and provide a foundation for subsequent requirements specification, system design, implementation, and testing. The gathered requirements are focused on creating a simple, efficient, and digitally managed restaurant ordering experience.